

Assignment 5 - Image and Video Generation Models

Applied Machine Learning Engineering

1 Your Challenge

In this assignment you will visually compare the effectiveness of two methods of manipulating images. The first method will put together several task-specific machine learning models you download from huggingface. The second method will use multipurpose online chat multimodal LLMs like copilot, chatgpt, or gemini, combining a text prompt with an image to manipulate.

2 Method 1 - Integrating Vision and Generative Models

Your challenge begins with a collection of natural images, some of which contain a squirrel, some of which contain a birdfeeder, some of which contain a squirrel eating at a birdfeeder, some of which contain only birds. Your first task is to build a program `subselectImages.py` which uses an object detection model to subselects only those images that contain a squirrel perched next to a birdfeeder, creating a textfile `selectedImages.txt` with the names of the selected files.

For each of the subselected images, your second task is to build a program `segmentSquirrels.py` that segments these images to build a mask around where the squirrel is, enlargening this mask a bit so that it is not too tight around the squirrel (for instance, accidentally leaving out a bit of its hair), saving the masks to image files.

In your third and fourth tasks, you will write programs that produce two sets of manipulated images using your subselected images and masks. The first program `removeSquirrels.py` uses a deep inpainting model on the selected images and masks with a neutral prompt to make it appear as if the squirrel was never there, saving the produced images. The second program `replaceSquirrelsWithBirds.py` replaces the squirrel with a bird.

Part of the score will test whether the images you selected were actually the ones with the squirrels perched at the birdfeeder. Part of your score will be a subjective assessment of how real each of your final fake images look. Hence, adjust your prompts, pads, selection code, and possible even the huggingface models utilized, away from the sample notebooks to improve these aspects. The details for what to submit and how it will be tested follow.

3 Method 2 - Using General Purpose Multimodal Online Assistants

Using at least 3 of the images your `selectImages.py` selected, use a online assistant of your choice to upload each image with a prompt you design requesting the squirrel to be replaced by a bird. Save the produced modified images `onlineModedN.jpg` (N being an integer counting across the different images you manipulate) and your chat transcripts links in `chatTranscriptLinks.txt`. Likewise use the online assistant of your choice (e.g. gemini's nano banana) on at least 3 of your images with a prompt you select to make it appear that the squirrel is not in the image, thereby creating the files `onlineErasedN.jpg` and a text file `eraseChatTranscriptLinks.txt` with links to your erasing chat transcripts. Part of your score will be a subjective assesment of how real each of your final fake images produced this way look, as well as the extent to which they leave the other contents of the photo intact. Hence, experiment with adjusting your prompts and which online AI assistant you utilize to help you do the manipulation.

3.1 Generating Images from Text Only

Using text prompts only in an online image generator of your choice, generate 3 convincing images of birds eating at birdfeeders, generating the image files `onlineGeneratedN.jpg` (N in 1,2,3) and the file `generationChatTranscriptLinks.txt`.

3.2 Generating Video

Pick a topic of your interest to generate an 8 second instructional video about using an online video generator like sora 2 or veo 3.1. Craft your prompt using a chat you share with link in `videoPromptGenerationChat.txt`, and your generated video as `myVideo.mp4`.

4 What to submit

Please submit a compressed archive `username-lab5.zip` containing the following files. For the shell commands below, you can assume that the transformers, diffusers, accelerate, etc huggingface packages will be installed in the environment in which they will be run.

1. subselectImages.py: The shell command

```
python subselectImages.py imageFileList outFileName
```

wherein imageFileList is the name of a text file containing, on each line, the path to a natural image file, should write a text file output named outFileName whose lines are those files containing a squirrel at a birdfeeder.

2. segmentSquirrels.py: The shell command

```
python segmentSquirrels.py sqImageFiles
```

wherein sqImageFiles is the name of a text file containing, on each line, the path to a natural image containing a squirrel at a birdfeeder, creates for each such image (suppose it is named imageName.jpg) a mask image (named imageName-sqMask.png and at the same path as imageName.jpg) giving a mask around the squirrel. This mask should have been enlarged a bit, to ensure that no traces of the squirrels hair are left outside the mask.

3. removeSquirrels.py: The shell command

```
python removeSquirrels.py sqImageFiles
```

should, after the segmentSquirrels command above has been run, and the masks have been generated, load each the file on each line of sqImageFiles (each of which contains a natural image of squirrel at a birdfeeder), and load the corresponding mask, then use a deep neural network inpainting model, together with a neutral prompt, to replace the area in the mask to make it appear as this was a natural image of only the birdfeeder with no squirrel there. The image created from imageName.jpg should be saved as imageName-squirrelRemoved.jpg.

4. replaceSquirrels.py: The shell command

```
python replaceSquirrels.py sqImageFiles
```

should, after the segmentSquirrels command above has been run, and the masks have been generated, load each the file on each line of sqImageFiles (each of which contains a natural image of squirrel at a birdfeeder), and load the corresponding mask, then, leveraging a stable diffusion inpainting model (together with, possibly segments of birds from other natural images), replaces the squirrel with a bird to generate a convincing fake image.

5. myImageFiles.zip: An archive containing:

- (a) selectedImages.txt the text file produced by subselectImages.py when run on my provided images,
- (b) your generated mask images,
- (c) your generated fake images with the squirrels removed,
- (d) and your generated images with the squirrels replaced with birds.
- (e) onlineModedN.jpg at least three images you generated with a multipurpose online AI assistant
- (f) onlineErasedN.jpg at least three images you generated to remove the squirrel using a multipurpose online AI assistant.
- (g) onlineGeneratedN.jpg at least 3 images you generated with text only prompts showing colorful birds at birdfeeders.
- (h) myVideo.mp4 the 8 second instructional video you made on a topic of your interest.

6. chatTranscriptLinks.txt the links to your chat transcripts with the online AI assistants to replace the squirrel.
7. eraseChatTranscriptLinks.txt the links to your chat transcripts with the online AI assistants to remove the squirrel.
8. generationChatTranscriptLinks.txt the links to your chat transcripts to generate convincing images of colorful birds at birdfeeder from text alone.
9. videoPromptGeneration.txt the chat in which you designed your video generation prompt using an AI assistant.

While I will provide you an original set of images to subselect from to develop with, I reserve the right to re-run your code with another set of images I have not shared.